

Builder Brief

Instructions for the coding agent that builds the system. Read Dossier 1 (System Specification) first — it is normative. This dossier tells you how to work, in which order, and what done looks like.

Version 1.0

Deliverables: codebase + AGENT.md + USER_SETUP.md + green evals

Method: small phases, each verified by running

1 · Mission and deliverables

Mission

Build the Trend Analyst system exactly as specified in Dossier 1, working like a senior engineer: plan first, build in small verified steps, leave the campsite cleaner than you found it. You are also the author of the two human/agent-facing documents the system needs to survive without you.

Deliverables (all four, no exceptions)

Deliverable	Acceptance bar
Codebase	Matches Dossier 1 layout; pytest green; ruff clean; mypy-strict on new modules; every module has tests with recorded fixtures
AGENT.md	Ops manual for the periodic monitor agent: persona, commands first, runbooks, three-tier boundaries, change classes (see section 5)
USER_SETUP.md	Every human action itemized: accounts, keys, approvals, cron, verification checklist, \$0 cost table (see section 6)
Evals green	50 golden cases in evals/cases.yaml; judge rubric and determinism checks passing in CI

2 · How to work (senior-engineer standards)

- **Read the spec first** — Dossier 1 is law. If spec and instinct conflict, follow the spec and flag the conflict to the human.
- **Track small tasks** — Keep a visible task list; exactly one item in progress; mark done only after verification ran.
- **Typed Python always** — Type hints on every function, Pydantic models at every boundary (config, plugin I/O, gate schemas).
- **Tests with every module** — No live network in tests — recorded fixtures with secrets redacted. A new source ships with its fixtures.
- **Verify by running** — After each phase: run pytest, the linter, and the real CLI path for what you built. Paste or log the evidence; never claim green from memory.
- **Small diffs** — One phase at a time, each independently working. Never rewrite two layers in one go.
- **No secrets, ever** — Nothing credential-like in code, logs, fixtures, or docs. If you almost commit one, stop and rotate the pattern, not just the value.
- **Stop on ambiguity** — Escalate to the human when: a spec invariant blocks you, an external API contradicts the docs, a destructive action is the only path, or evals and judgment disagree twice in a row.

3 · Phase plan (build in this order)

P0 — Skeleton, config, registry, health

Repo layout per spec section 3. Layered settings (defaults < YAML < env < CLI), sources.yaml loader with Tier-A-in-L0 hard-fail validation, Postgres models plus first migration, health CLI printing stub statuses. DONE WHEN: pytest green, ruff clean, health runs and reports an empty-but-healthy system.

P1 — L0 collectors plus ledger (3 Tier-S sources)

Implement HN Firebase, Wikipedia Pageviews, and one more Tier-S source as SourcePlugins with token buckets, watermarks, content-hash dedup, and run ledger writes. Crash-recovery test: kill mid-run, rerun, assert only unfinished work executes. DONE WHEN: two consecutive nightly-equivalent runs, second run near-zero new work, ledger proves it.

P2 — L1 mining plus scoring v1

Phrase extraction, per-category EWMA z-scores, 95 percent prune, MGS sub-scores in code, fad features, Monte Carlo revenue triple. Seed 10 golden eval cases from real outputs. DONE WHEN: ranked table renders end-to-end from L0 data with snapshots versioned.

P3 — LLM gates, cache, gateway

Gateway with Gemini -> Groq -> Cerebras -> local fallback, per-gate budgets, 30-day cache, Pydantic gate schemas, code-enforced grounding rule. Start with mocked provider responses, then exactly one live run to prove failover and accounting. DONE WHEN: judge and writer produce schema-valid, cited outputs on golden inputs; token log shows spend per gate.

P4 — L2 enrichment, briefs, snapshots

eBay Browse and one more Tier-A plugin (L2 only, on-demand, top-K), brief rendering, snapshot history views, TTL jobs. DONE WHEN: full nightly run completes inside budget with quota ledger balanced to zero unexplained spend.

P5 — Documents, evals, monitor dry-run

Write AGENT.md and USER_SETUP.md to the bars in sections 5 and 6. Grow evals to 50 cases, wire CI (tests, evals, determinism). Simulate the five incident runbooks against a staging copy and record the monitor agent handling them. DONE WHEN: section 8 checklist is fully ticked and the human has signed the verification steps.

Order is load-bearing. No LLM work before P2 scoring exists (gates need real inputs). No L2 before budgets and ledger exist (quota needs a leash). No documents before the system they describe exists (docs describe reality, not wishes).

4 · Agent-engineering rules (distilled, enforced)

Drawn from current production practice (Anthropic 2025–2026: simple composable patterns, evaluator-optimizer, orchestrator-workers, LLM-judge evals; LangGraph 2026: checkpoint-every-transition, replay determinism, ledgers outside the rollback surface). Applied here as:

- **Workflows over agents** — The pipeline is fixed-path code, not an agent loop. LLM calls are single-shot gates with schemas.
- **Orchestrator-workers only where subtasks are unpredictable** — Permitted inside the Writer gate (angles per category). Nowhere else in v1.
- **Evaluator-optimizer for briefs** — Writer drafts, a narrow critic pass checks rubric items (citations, ranges not points, risks present), one revision max.
- **Tools are UI** — Each plugin and each gate tool gets one purpose, a clear description, an example, and its budget in code. If two tools overlap, merge them.
- **Scale effort to complexity** — Hard caps everywhere: max candidates per gate, max enrich calls per night, max one Writer revision. Effort budgets live in config, not prompts.
- **Eval before features** — A behavior change without a golden case proving the old behavior is a regression risk, not progress.
- **Checkpoint plus external ledger** — Resume-safe by construction (spec section 5). Test the kill-and-resume path, not just the happy path.
- **Human interrupt for the irreversible** — Anything destructive, spend-increasing, or schema-changing proposes and waits. The monitor agent inherits this (AGENT.md).
- **Context hygiene** — Tool outputs are truncated with sensible defaults; state stores references and hashes, never blobs. Prompts receive slices, not lakes.

5 · AGENT.md requirements (you MUST write this file)

AGENT.md is the operating manual for the periodic monitor agent. It MUST open with executable commands (health, tests, evals, common fixes with flags), then project structure and stack with versions, then the three-tier boundary system below. Keep it under ~150 lines; link out for detail.

5.1 Boundary tiers (copy this shape, fill with real paths)

Tier	Contents
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Always do	Read health output first; append to logs; restart a failed run from checkpoint; clear stale cache entries older than TTL; report what you did
Ask first (file a proposal, wait)	Enable/disable a source; change thresholds, weights or budgets; add dependencies; modify CI schedule; touch migrations
Never do	Commit secrets; delete snapshots, ledger rows or eval cases; edit .env or production config; force-push; bulk-delete raw lake rows; approve your own proposal

5.2 Required sections

- Role and persona: careful SRE junior, evidence-first, proposes before acting outside the safe list.
- Monitor loop procedure: read health, classify (healthy / degraded / down), match a runbook, act within tier, verify, log, escalate with the exact template.
- Five runbooks minimum: HTTP 429 storm from a source; source approval/schema change breaking a plugin; quota burn above 80 percent; judge keep-rate drift outside band; eval baseline drop.
- Change classes with examples: SAFE (restart from checkpoint), APPROVAL (disable flaky source for 7 days), FORBIDDEN (manual snapshot edits) — each with verification step.
- Escalation template: symptom, evidence (commands plus output), attempted safe actions, proposed approval-class change, blast radius if ignored.
- Rollback rule: every change you make **MUST** be reversible in one command, stated in the log line.

6 · USER_SETUP.md requirements (you **MUST** write this file)

USER_SETUP.md is the human action list. The human is non-technical here: every item needs where to click, what to paste where, what it costs, and how to verify. Structure:

- Key table with one row per credential: purpose, where to create it (exact console URL pattern), free tier limit, where to put it (.env key name), verify command and expected output.
- Accounts and approvals to file on day one, with expected wait times and what to do meanwhile (build never blocks on approvals: sources start disabled).
- Scheduler setup: the exact cron entry or Actions workflow to enable, plus how to trigger a manual run.
- First-run checklist in order, ending with the health command and what healthy output looks like.
- Cost summary table proving \$0/month at expected volume, per provider, with the cap that would trigger each paid tier.
- Troubleshooting: the five most likely setup failures (bad key, wrong region, OAuth expiry, quota zero-day, Postgres connection) with fix per failure.

Test of truth. If the human cannot go from zero to a green health check using only USER_SETUP.md, the document — and therefore your build — is not done.

7 · Anti-patterns (any of these fails review)

- Framework-first: importing an agent framework before the plain-code pipeline works.
- LLM in the hot loop: any gate seeing unpruned L0 volume.
- Tier-A in L0, or any network call without timeout, retry, budget and ledger entry.
- Overwriting snapshots or mutating history to fix a bug — append a correction, never rewrite.
- Prompt-only budgets or unenforced grounding (“the prompt says to cite”).
- Live network in tests, secrets in fixtures, unpinned dependencies.
- Silent failure: an exception path that logs nothing and continues as if healthy.
- A swarm of agents where a checklist would do. Complexity **MUST** cite a measured need.

8 • Definition of done

- pytest green, ruff clean, mypy-strict on new modules — evidenced by command output, not memory.
- Full pipeline runs end-to-end on real sources within nightly budgets; quota ledger balances.
- Kill-and-resume proven for L2; replay determinism hash-matches in CI.
- 50 golden eval cases green; no rubric down more than 2 points from baseline.
- AGENT.md and USER_SETUP.md exist and meet sections 5–6 bars; monitor dry-run against staging logged.
- Human runs the USER_SETUP.md checklist alone and reaches green health. Signature recorded in the repo log.

End of Dossier 2. Build P0 first. Verify everything by running. Leave the campsite cleaner than you found it.